

Unraveling the epidemiological and dispersal dynamics of the 2024–2025 chikungunya virus epidemic on Réunion Island

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PMID: 42748151 • DOI: <https://doi.org/10.1073/pnas.2621019123> • Repository: https://github.com/sdellicour/chikungunya_reunion

Abstract

Between 2024 and 2025, Réunion Island experienced a major chikungunya virus (CHIKV) epidemic resulting in over 54,000 confirmed clinical cases, occurring approximately twenty years after the devastating 2005–2006 epidemic. Here, we analyze a comprehensive genomic dataset of viral sequences collected during the epidemic to reconstruct the spatiotemporal introduction and dispersal history of CHIKV across the island. Using time-scaled phylogenetic and discrete phylogeographic inferences in BEAST 1.10.5 coupled with BEAGLE GPU acceleration, we demonstrate that the outbreak originated from a single introduction event and spread following a gravity-model dynamic, where viral lineage transitions occurred preferentially between high-density human population centers. Generalized linear model (GLM) extensions of sampling-aware coalescent skygrid models reveal statistically significant associations between transmission intensity and climatic covariates, specifically monthly mean temperature and cumulative precipitation. Finally, mathematical modeling indicates that accumulated population immunity from both the historic 2005–2006 and 2024–2025 epidemics is sufficient to explain epidemic curtailment, suggesting limited vulnerability to near-term large-scale resurgence.

BEAST 1.10 Benchmark Specification

Parameter	Publication / BEAST Specification	ChronAeon Comparator
Taxa & Length	251 taxa, 11,172 bp (CHIKV complete genome)	Identical MSA parity (N=251, L=11,172 bp)
Clock Model	Uncorrelated Lognormal Relaxed Clock (UCLN)	AutoClock (K*=3 multi-rate communities)
Tree Prior	Bayesian Skyline / Sampling-aware Skygrid Coalescent	ChronAeon Penalized Spline Rate-smoothing
Evolutionary Rate	8.8×10^{-4} subs/site/year (95% HPD: $6.2\text{--}11.6 \times 10^{-4}$)	2.52×10^{-4} subs/site/year (Community 2: 7.9×10^{-4})
tMRCA Inference	2004.8 CE (95% HPD: 2004.5–2005.1 CE) for crown	1939.27 CE (Stem ancestor of ECSA lineage)
Computation Time	2 to 8 weeks on high-performance GPU cluster	8.86 seconds (instantaneous single-core)